THE CONTENT IDEAL OF AN ELEMENT OF A FLAT MODULE IS FINITELY GENERATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Content ideal of an element of a flat module. definition environment at source lines 5699-5704 of the same chapter, immediately before the counted statement. Supplies the object of the statement, namely the smallest ideal I with x in IM, and the hypothesis that it exists.
- The set of ideals I with x in IM is closed under finite sums and contains (f_1, ..., f_n) whenever x is such a finite combination. used at proof lines 5719-5720 together with the flatness of M. Produces the finitely generated candidate ideal (f_1, ..., f_n) inside I.
- Minimality of the content ideal. used at proof line 5720-5721, from the defining property of the content ideal. Turns the inclusion of the finitely generated ideal into I into the statement that I itself is finitely generated.

Source reference. Original lemma 0ASB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13574–13574. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let A be a ring. Let M be a flat A-module. Let $x \in M$. The content ideal of x, if it exists, is finitely generated.*

Proof. Say $x \in IM$. Then we can write $x = \sum_{i=1,\ldots,n} f_i x_i$ with $f_i \in I$ and $x_i \in M$. Hence $x \in I'M$ with $I' = (f_1, \ldots, f_n)$. □

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.